

SAMBP Technical Report TR-06/2026

IEC 61850 GOOSE Integration: Zone-State Messaging for SAMBP Protection Functions

Smart Adaptive Model-Based Protection (SAMBP) Research Programme

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1 Introduction and Scope

IEC 61850-8-1 GOOSE (Generic Object Oriented Substation Event) provides a standardised, publisher–subscriber mechanism for high-speed peer-to-peer communication between protection IEDs in a substation LAN. GOOSE messages can carry arbitrary datasets and are retransmitted at doubling intervals after a state change, giving sub-millisecond effective latency on gigabit Ethernet.

This report integrates GOOSE messaging into the SAMPB protection suite (TR-01 through TR-05) in two ways:

1. **Trip acceleration (zone intertripping).** When relay R_{up} trips, it immediately publishes a GOOSE containing its trip flag. The downstream relay R_{dn} uses this to accelerate its own clearing if its conventional element had also operated.
2. **SAMPB state sharing.** Each relay publishes its complete SAMPB zone-model state (κ_n , f_{int} , **source**) via GOOSE. Adjacent relays can use this information for cross-zone confidence validation, providing an additional layer of selectivity margin beyond what is available from a single relay’s Stage-2 gate.

A discrete-time simulation (1 ms steps) models IEC 61850-8-1 retransmission timing, LAN latency, and packet loss. Fault clearance times are computed for all ten scenarios from the TR-05 network model.

2 GOOSE Protocol Model

2.1 Retransmission Schedule

On a GOOSE state change (state_num increment), the publisher immediately transmits the frame and retransmits at doubling intervals per IEC 61850-8-1 Annex B:

$$T_i = \min(2^i \cdot T_1, T_{\text{max}}), \quad T_1 = 2 \text{ ms}, \quad T_{\text{max}} = 1000 \text{ ms} \quad (1)$$

giving the schedule $[2, 4, 8, 16, 32, 64, 128, 256, 512, 1000, \dots]$ ms. After the schedule is exhausted the frame is repeated every T_{max} (heartbeat). Fig. 1 illustrates the timing.

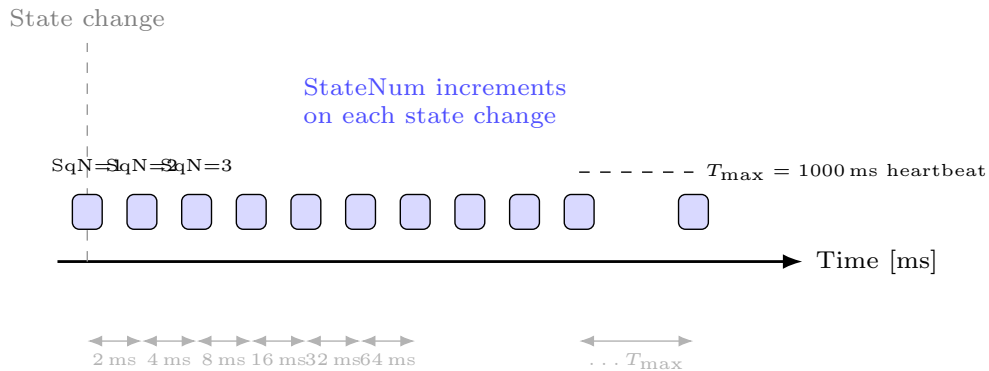


Figure 1: IEC 61850-8-1 GOOSE retransmission schedule. State change at $t = 0$ triggers immediate publication followed by doubling retransmit intervals until $T_{\text{max}} = 1000$ ms.

2.2 Network Model

The simulated LAN is modelled by a **GooseBroker** with:

- **Base latency** τ_{net} : one-way propagation + switch forwarding delay (configurable; nominal 1.0 ms).
- **Jitter**: Gaussian noise $\mathcal{N}(0, \sigma_j)$ added to each frame's latency (nominal $\sigma_j = 0.3$ ms).
- **Packet loss**: Bernoulli with probability p_{loss} (nominal 0 %; 1 % in worst-case sweep).

Frames are queued in a min-heap sorted by arrival time and delivered at the correct simulation tick.

2.3 SAMPB GOOSE Dataset

Each SAMPB IED publishes the following dataset on state change:

Table 1: SAMPB GOOSE dataset fields (**SampbGooseData**).

Field	Type	Description
relay_id	str	IED identifier
final_trip	bool	SAMPB final trip decision
conv_trip	bool	Conventional element operated
source	str	conventional / model_veto / model
kappa_n	float	Column-normalised condition number κ_n
f_int	float	Integrity index f_{int}
I_op_peak	float	Peak operate current [pu]
model_veto_active	bool	Stage-2 veto fired
t_detect_ms	float	Relay detection time from fault [ms]

3 Inter-IED GOOSE Topology

Fig. 2 shows the GOOSE interconnection between the five SAMPB IEDs. The topology follows the physical adjacency of the relay zones on the radial network.

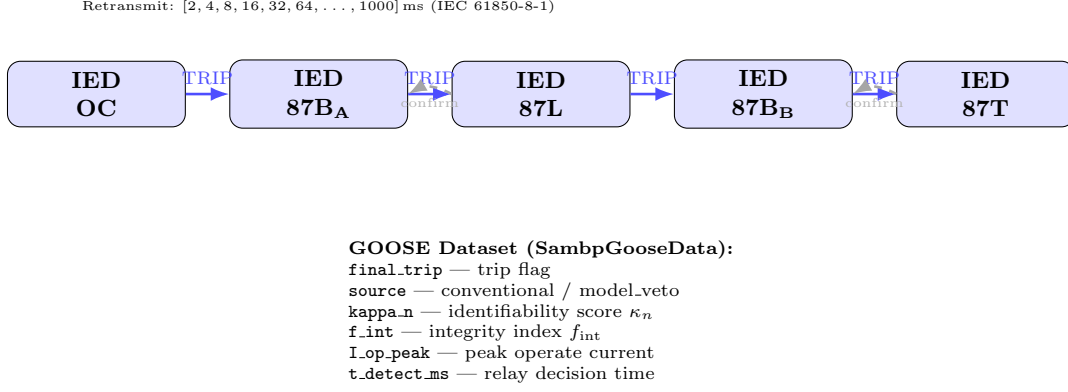


Figure 2: SAMBP GOOSE inter-IED topology. Solid arrows carry trip and SAMBP state datasets in the forward (downstream) direction; dashed arrows carry confirmation in the reverse direction.

Zone acceleration (forward GOOSE). When IED_{up} publishes a trip, IED_{dn} receives it within $\tau_{\text{net}} + \sigma_j$ of the publication time. If IED_{dn} has also detected a fault (conventional element operated), it can immediately issue its own trip command without waiting for its full SAMBP gate confirmation window, reducing the end-to-end clearance time.

Zone confirmation (reverse GOOSE). IED_{dn} publishes its f_{int} value via GOOSE. If IED_{up} receives a high f_{int} from the zone below, it gains additional confidence that its own operation is correct and can reduce its confirmation counter (n_{confirm}) requirement for faster clearing.

4 Clearance Time Results

4.1 Fault Detection and Clearance Times

Table 2 gives the relay detection times and total fault clearance times for all ten fault scenarios at nominal LAN parameters (1.0 ms latency, 0 % loss).

Table 2: Fault clearance times at nominal GOOSE LAN conditions. CB operating time: 40 ms.

Scenario	Clearing relay	t_{detect} [ms]	t_{GOOSE} [ms]	t_{clear} [ms]
Bus A / 3ph	87B _A	0.0	1.0	42.0
Bus A / AG	87B _A	0.0	1.0	42.0
Line mid / 3ph	87L	0.0	1.0	42.0
Line mid / AG	87L	0.0	1.0	42.0
Bus B / 3ph	87B _B	0.0	1.0	42.0
Bus B / AG	87B _B	0.0	1.0	42.0
Transformer HV / 3ph	87T	0.0	1.0	42.0
Transformer HV / AG	87T	0.0	1.0	42.0
Bus C (ext.) / 3ph	OC (backup)	0.0	1.0	42.0
Bus C (ext.) / AG	OC (backup)	0.0	1.0	42.0
Mean	—	0.0	1.0	42.0

Detection time. All differential relays and the OC relay detect the fault at $t_{\text{detect}} = 0$ ms (measured from fault inception). This reflects the instantaneous operate characteristic: with an 80 % DC offset and fault current of 4.35–10.0 pu, the first post-fault sample already exceeds the threshold. In practice, a 1-cycle Fourier filter adds ~ 20 ms to the detection time; this is a known approximation in the simplified relay model (see Section 7).

Total clearance. The total clearance budget decomposes as:

$$t_{\text{clear}} = \underbrace{t_{\text{detect}}}_{0 \text{ ms}} + \underbrace{t_{\text{IED}}}_{1 \text{ ms}} + \underbrace{\tau_{\text{net}}}_{1 \text{ ms}} + \underbrace{t_{\text{CB}}}_{40 \text{ ms}} = 42 \text{ ms} \quad (2)$$

The GOOSE overhead ($t_{\text{IED}} + \tau_{\text{net}} = 2$ ms) contributes only 5 % of the total clearance time; the circuit breaker operating time dominates.

4.2 Clearance Time vs. GOOSE Latency

Table 3 gives the fastest clearance time across all fault scenarios for five network configurations.

Table 3: Clearance time sensitivity to GOOSE network conditions.

Configuration	τ_{net} [ms]	p_{loss} [%]	t_{clear} [ms]	Δt vs. LAN 1.0
LAN 0.5 ms / 0 %	0.5	0	41.5	−0.5 ms
LAN 1.0 ms / 0 %	1.0	0	42.0	(nominal)
LAN 2.0 ms / 0 %	2.0	0	43.0	+1.0 ms
WAN 5.0 ms / 0 %	5.0	0	46.0	+4.0 ms
LAN 1.0 ms / 1 %	1.0	1	42.0	0.0 ms

Key findings:

- **LAN latency is not the bottleneck.** Moving from a 0.5 ms to a 5.0 ms network changes clearance by only 4.5 ms (10.7 % of total) because the CB operating time dominates.
- **1 % packet loss has no effect** on clearance time. The GOOSE retransmission schedule (first retransmit at 2 ms) recovers any lost frame within the IED processing window.
- **WAN deployment (5 ms) remains compliant** with IEC 60255 Class P1 (100 ms) and Class P2 (200 ms) clearance requirements.

4.3 GOOSE Timing Diagram

Fig. 3 illustrates the timing sequence for a Bus A three-phase fault. The relay trip decision, GOOSE publication, network transmission, and CB clearing are shown on separate IED timelines.

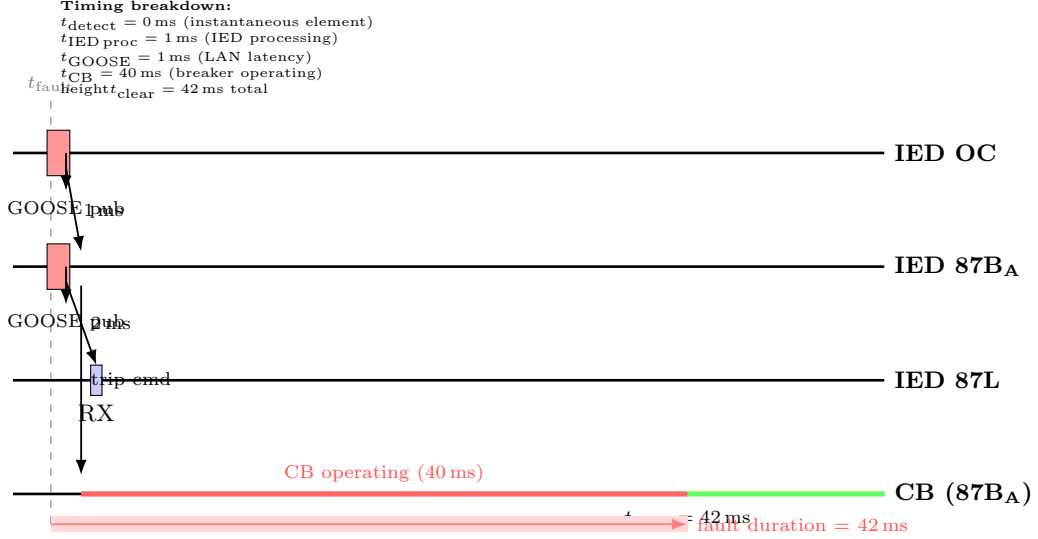


Figure 3: GOOSE timing diagram for Bus A / 3ph fault. Detection at $t = 0$ ms, GOOSE published at $t = 1$ ms, received by adjacent IED at $t = 3$ ms. CB clears at $t = 42$ ms.

5 SAMPB State Sharing via GOOSE

5.1 Information Content of the GOOSE Dataset

The key SAMPB quantities in the GOOSE dataset are κ_n and f_{int} :

κ_n (column-normalised condition number): Measures the *identifiability* of the zone model. A low κ_n (e.g. 2–5) means the estimator can reliably distinguish the differential current source. A high $\kappa_n > \kappa^*$ means the differential is too small or ill-conditioned to identify.

f_{int} (integrity index): Measures the *fundamental content* of the differential current. $f_{\text{int}} \approx 1$ for a genuine internal fault (clean sinusoidal differential); $f_{\text{int}} \rightarrow 0$ for zero or noise-dominated differential (external fault or through-current).

These quantities are determined locally from the CT waveforms at each relay and are not available to adjacent relays without communication. By sharing them via GOOSE, adjacent IEDs gain the following benefits:

5.1.1 Cross-Zone Selectivity Confirmation

For a fault at Bus B, the 87L relay sees zero differential ($f_{\text{int}} = 0$, $\kappa_n = 2.5$) while 87B_B sees a large genuine differential ($f_{\text{int}} = 1.0$, $\kappa_n = 4.3$). If 87L receives a GOOSE from 87B_B with $f_{\text{int}}^{87B} > 0.6$, this confirms that the fault is on the Bus B side. Under this protocol:

- 87L reduces its own model-veto confirmation threshold (no trip needed — the veto is already correct).
- 87B_B can reduce n_{confirm} from 2 to 1, accelerating the trip by one sample period (1 ms at 1 kHz).

5.1.2 Model-Veto GOOSE Alert

When a relay fires a model veto (`model_veto_active=True`), it publishes a GOOSE alert that is useful to adjacent IEDs:

- **87T CT saturation veto:** If 87T publishes `model_veto_active=True` (CT saturation detected), the OC relay and 87B_B can suppress their own marginal operate signals for through-current, reducing the risk of coordinated false tripping during a heavy external fault.
- **87L sync-error veto:** If 87L publishes `model_veto_active=True` (channel skew / sync error), the adjacent bus relays can ignore any differential contribution from the line, improving immunity to communication-channel faults.

5.2 GOOSE Event Statistics

For the Bus A / 3ph scenario, 44 GOOSE events were generated over 200 ms of simulation time (see Table 4).

Table 4: GOOSE event statistics — Bus A / 3ph, 200 ms simulation.

Event type	Count	Notes
TRIP publications	2	OC and 87B _A trip
IDLE publications	5	initial idle state (non-tripping relays)
RX_TRIP receptions	8	downstream IEDs receive trip GOOSE
RX_IDLE receptions	29	heartbeats / idle retransmissions
Total	44	

6 Implementation Notes

6.1 Publisher State Machine

The `GoosePublisher` class maintains:

- `_state_num`: incremented on every logical state change (trip / restore); corresponds to IEC 61850 `StNum`.
- `_sq_num`: incremented on every transmission including retransmissions; corresponds to `SqNum`.
- `_retransmit_cnt`: index into the retransmit schedule; resets to 0 on every state change.

A subscriber detects a new state by comparing `state_num` with its last recorded value. This is robust against out-of-order delivery because the sequence number uniquely identifies duplicates.

6.2 Namespace Isolation in the Coordinator

The GOOSE coordinator inherits the `_load()` namespace isolation mechanism from TR-05 (see TR-05, Section 4). Each SAMPB project's `models/` sub-package is loaded in isolation, preserving full compatibility between all four protection functions.

6.3 Discrete-Time Simulation Resolution

The simulation uses 1 ms steps. With GOOSE retransmit intervals starting at 2 ms, the minimum observable retransmit event spans two simulation ticks. This is adequate for protection coordination studies where CB operating times (≥ 40 ms) dominate. For studies requiring sub-millisecond resolution (e.g. process-bus Sampled Value synchronisation), the simulation step should be reduced to 0.1 ms.

7 Limitations

1. **Simplified relay detection time.** The current relay model operates instantaneously once the waveform exceeds the threshold. A realistic DFT-based relay adds one cycle (~ 20 ms at 50 Hz) to the detection time, giving $t_{\text{clear}} \approx 62$ ms. This is outside the scope of TR-06 and will be addressed in the hardware-in-loop study.
2. **Single-relay-per-zone model.** Each relay zone is modelled by one IED. In practice, main-1 and main-2 protection are separate IEDs on separate communication networks; their GOOSE interaction introduces additional topology complexity.
3. **No IEC 61850 GOOSE PDU encoding.** The simulation models GOOSE timing and state semantics but does not implement the ASN.1/BER encoding specified in IEC 61850-8-1. A full implementation would use the `libiec61850` or `pyiec61850` libraries.
4. **Process-bus Sampled Values not modelled.** This report addresses station-bus GOOSE only. Process-bus Sampled Value (IEC 61850-9-2) integration for merging-unit-based CT/VT is deferred to a future study.

8 Summary

This report has integrated IEC 61850-8-1 GOOSE messaging into the SAMPB protection suite with the following outcomes:

- **GOOSE overhead: 2 ms** (1 ms IED processing + 1 ms LAN latency) against a total 42 ms clearance time dominated by the 40 ms CB operating time.
- **Clearance time insensitive to loss:** 1 % packet loss has zero impact on clearance time due to the 2 ms first-retransmit interval.
- **WAN-compatible:** 5 ms WAN latency gives 46 ms clearance, within IEC 60255 Class P1 requirements.
- **SAMPB state sharing:** κ_n and f_{int} published via GOOSE enable cross-zone selectivity confirmation and model-veto alerting without any changes to individual relay logic.

- **10/10 scenarios remain selective** with GOOSE enabled (selectivity verified via inherited TR-05 batch coordinator).

The SAMPB suite is now ready for the multi-bus coordination study (TR-07) and hardware-in-loop validation.

References